

ENTRY 021 Six-Edge Protocol Test on FakeKyiv — August 4, 2026 — Status: COMPLETE

Entry 021 — The Forgiveness Law Survives. The Measurement Protocol Does Not.

Six Edges on FakeKyiv (127 Qubits) Expose a Decoherence Confound in Entries 017–020

1. What Was Done

Entry 020 closed by proposing a third-chip test of the ~0.63 ratio-of-ratios finding. This entry runs that test on **FakeKyiv** (127 qubits, heavy-hex, *ecr* native gate) using the same cancellation-based decay-curve method as Entries 017, 019, and 020 — but with one deliberate change to the protocol: **six edges were measured instead of two**, spanning 0.31% to 4.32% raw calibrated gate error.

That single change was enough to invalidate the headline finding of Entry 020 and to expose a confounding variable present in the Entry 019 measurement that the two-point method was structurally incapable of detecting.

2. Why Two Points Were Never a Test

Entries 017/019 (Cairo) and Entry 020 (Sherbrooke) each measured exactly two edges per chip — one low-error, one high-error — and drew a power law through them. **Two points define a power law exactly, with zero residual.** The fit could not fail, so it never tested anything. It reported the two measurements back in a different coordinate system.

The ratio-of-ratios agreement celebrated in Entry 020 (Cairo 0.6302, Sherbrooke 0.6306) is also weaker than it appeared. The two chips were probed over *different error spans* — 5.2x for Cairo, 8.8x for Sherbrooke. Under a genuine shared power law, a wider span must produce a *smaller* high/low ratio. Matching ratios across different spans therefore imply **different exponents**, and they do: –0.280 for Cairo against –0.212 for Sherbrooke. The agreement was arithmetic coincidence, not physics.

3. Chosen Edges

Edge	Category	Raw calibrated error	Selection rationale
Kyiv (119,120)	very low	0.3113%	chip minimum
Kyiv (28,29)	low	0.6013%	matches Cairo's low edge (0.60%)
Kyiv (77,78)	mid-low	0.9860%	fills the decade gap
Kyiv (45,46)	mid	1.6181%	chip 75th percentile
Kyiv (66,67)	high	2.6181%	near Cairo / Sherbrooke high edges
Kyiv (23,24)	very high	4.3229%	highest plausible calibrated edge

Nine gate counts per edge (1 to 41 CX), 4096 shots each, seed 1729. Placeholder calibration entries (gate error ≥ 50%) were filtered out as in Entry 020.

4. Results — The Six-Point Fit Fails

Fitted across all six edges:

Edge	Raw error	Measured ratio	Fitted	Residual
(119, 120)	0.3113%	0.4345	0.8076	–0.3731
(28, 29)	0.6013%	1.5904	0.6497	+0.9406
(77, 78)	0.9860%	0.5353	0.5518	–0.0166
(45, 46)	1.6181%	0.4308	0.4685	–0.0377
(66, 67)	2.6181%	0.3937	0.3997	–0.0060
(23, 24)	4.3229%	0.2927	0.3386	–0.0460

ratio = $0.1200 \times \text{err}^{-0.3303}$ $R^2 = 0.2956$ — a failed fit. Four edges sit almost exactly on the curve; two are wildly off. Edge (28,29) returns a forgiveness ratio of **1.5904**, which is not merely a large error but *physically impossible* under the model: it says the pair decayed faster than its own gate error permits. Something other than the two-qubit gate was driving that decay.

5. Diagnosis — Decoherence, Not Gate Error

The forgiveness ratio is defined as *measured decay constant* ÷ *raw gate error*. That definition silently assumes the decay is *caused* by gate error. But every CX also occupies finite time — 562 ns on Kyiv — during which the qubits dephase. Where T2 is short relative to gate duration, the measured decay is dominated by decoherence and the ratio stops measuring forgiveness altogether.

A domination factor makes this testable: **$D = \text{raw gate error} \div (\text{gate duration} \div T2_{\min})$** . $D > 1$ means gate error dominates; $D < 1$ means decoherence does.

Edge	Raw error	T2 min	Decoherence / gate	D	Verdict
(119, 120)	0.3113%	202.7 μs	0.277%	1.12	contaminated
(28, 29)	0.6013%	12.7 μs	4.430%	0.14	contaminated
(77, 78)	0.9860%	212.3 μs	0.265%	3.73	usable
(45, 46)	1.6181%	57.2 μs	0.983%	1.65	usable
(66, 67)	2.6181%	86.6 μs	0.649%	4.03	usable
(23, 24)	4.3229%	65.1 μs	0.863%	5.01	usable

Qubit 28 carries a **T2 of 12.7 μs** against a chip median in the hundreds — roughly seven parts dephasing to one part gate error. It was never measuring forgiveness. Edge (119,120) sits at $D = 1.12$, where the two mechanisms are comparable, which depresses its ratio below trend.

Refitting on the four edges that satisfy $D \geq 1.5$:

ratio = $0.0898 \times \text{err}^{-0.3873}$ $R^2 = 0.9648$

R^2 moves from 0.296 to 0.965 on removal of two points. **The power-law form was never the problem. The edge-selection protocol was.** Edges must be screened on D, not on error magnitude alone.

6. Retroactive Screen — Entry 019's Anchor Is Contaminated

Applying the same screen backwards to every edge used in Entries 017–020:

Chip	Edge	Raw error	T2 min	D	Verdict
Cairo	(24, 25)	0.5992%	192.4 μs	4.32	usable
Cairo	(19, 22)	3.1315%	6.9 μs	0.31	CONTAMINATED
Sherbrooke	(60, 61)	0.3470%	360.3 μs	2.34	usable
Sherbrooke	(66, 73)	3.0667%	170.0 μs	9.78	usable

Cairo's high-error edge (19,22) has **T2 = 6.9 μs** and a 697 ns gate — **10.1% decoherence per gate against 3.13% gate error**. Entry 019's measured ratio of 0.334 at that edge is roughly three parts dephasing to one part gate error.

That number is one of the two points hardcoded in `emulator_v3_routing.py` :: `variable_forgiveness_ratio()`. **The emulator's error model is currently anchored on a broken qubit.** Entry 020's two Sherbrooke edges both pass the screen; that entry's measurements stand.

7. Impact on the Emulator

Raw edge error	Old ratio (Cairo-anchored)	New ratio (Kyiv, 4 clean edges)	Change
0.300%	0.6433	0.8518	+32.4%

Raw edge error	Old ratio (Cairo-anchored)	New ratio (Kyiv, 4 clean edges)	Change
0.500%	0.5577	0.6989	+25.3%
1.000%	0.4595	0.5343	+16.3%
2.000%	0.3785	0.4085	+7.9%
3.000%	0.3380	0.3491	+3.3%
4.500%	0.3018	0.2984	−1.1%

The two laws agree near 3–4.5%, where the contaminated point pinned them together, and diverge sharply on low-error edges — which are the majority of any chip. **75% of Kyiv's calibrated edges sit below 1.6%.**

Circuit-level consequence, N CNOTs on a 1.0% edge:

CNOTs	Old prediction	New prediction	Gap
1	99.54%	99.47%	−0.07 pts
10	95.50%	94.78%	−0.72 pts
20	91.20%	89.84%	−1.36 pts
40	83.18%	80.71%	−2.46 pts

The current emulator is **optimistic on low-error edges**, and the error compounds with depth. At 40 CNOTs the correction is 2.46 points — comparable in size to the 1.87-point validation gap that Entry 019 recorded as a success.

8. Robustness Check

Before accepting two outliers as real, the ratios were re-measured across three simulator seeds:

Edge	Seed 1729	Seed 42	Seed 20260804	Spread
(77, 78)	0.5353	0.5583	0.5330	0.0253
(23, 24)	0.2927	0.3076	0.3143	0.0216

Seed spread is ~0.02. The two outlier residuals are 0.37 and 0.94 — fifteen to forty times larger. The outliers are physics, not sampling noise.

9. What This Means for QuantumBridge

Edge screening becomes part of the protocol. Any edge used to fit a forgiveness ratio must satisfy $D \geq 1.5$, and D must be reported alongside every future measurement. This is cheap — it reads T2 and gate duration straight from calibration data, with no extra circuits.

Entry 020's calibration shortcut is withdrawn. The proposal to calibrate a new chip from its low-error baseline plus a portable −0.63 constant does not survive. Kyiv gives 0.674 over a 13.9x span, and the Cairo/Sherbrooke agreement rested on one contaminated point. The cost saving was real but the constant underneath it was not.

Chip-invariance of the exponent remains unproven. Kyiv's clean four-point −0.387 still differs from Sherbrooke's two-point −0.212. Sherbrooke and Cairo need the same six-edge treatment before any cross-chip claim is defensible.

Honest caveat. All of this is measured against Qiskit's FakeKyiv noise model, not live hardware — IBM Quantum Platform access lapsed with the trial account on August 4. The $T_2 = 12.7 \mu\text{s}$ on qubit 28 is a genuine recorded calibration value from a real device snapshot and the decoherence mechanism is real, but the *magnitude* of the correction should be re-validated on hardware when access is restored. The protocol change holds regardless.

Entry 021 — Conclusion

Widening the measurement from two edges to six turned an untestable fit into a failing one, and the failure was informative. Two of six Kyiv edges produced forgiveness ratios inconsistent with the power law, including one physically impossible value of 1.59; both were traced to qubits whose T2 is short enough that decoherence, not gate error, drives the observed decay. Screening edges on the domination factor D and refitting the four clean points

restores the power law at $R^2 = 0.9648$. Applying the same screen retroactively shows that the high-error anchor point of Entry 019 — currently hardcoded into the v3 emulator — fails it, making the shipped error model optimistic by up to 32% on low-error edges. The ratio-of-ratios finding of Entry 020 is withdrawn. The forgiveness effect itself is unharmed; what changed is that it now has a stated validity condition and a fit with enough points to have residuals at all.

10. Next Step — Entry 022

Re-run the six-edge, D-screened protocol on Sherbrooke and Cairo so all three chips are compared on equal footing, then re-anchor `variable_forgiveness_ratio()` and re-validate Emulator v3 against the Bell and GHZ circuits from Entry 019. A secondary question worth testing: whether the residual exponent differences track chip generation, since Cairo is an older Falcon-class device while Sherbrooke and Kyiv are both Eagle-class.

Research Log — Updated Index (Entries 017–021)

Second addendum to the Table of Contents. Page 2 lists Entries 001–012 and page 4 lists Entries 013–016; this page extends the index rather than replacing either.

Entry	Title	Date	Status
Entry 017	Empirical Forgiveness Ratio from Gate-Count Decay Curve	Aug 1, 2026	Complete
Entry 018	GHZ Decay Curve — Forgiveness Beyond Bell Circuits	Aug 2, 2026	Complete
Entry 019	Error-Magnitude-Aware Forgiveness (Validation Gap 6.14 → 1.87 pts)	Aug 4, 2026	Superseded
Entry 020	Cross-Chip Generalization Test — FakeSherbrooke	Aug 4, 2026	Partly withdrawn
Entry 021	Six-Edge Protocol Test — Decoherence Confound Found	Aug 4, 2026	Complete

Entry 019 is marked *Superseded* because its high-error anchor point fails the $D \geq 1.5$ screen introduced here; its low-error measurement stands. Entry 020 is marked *Partly withdrawn*: its core finding that forgiveness constants are chip-specific holds, while the ratio-of-ratios constant it proposed does not.